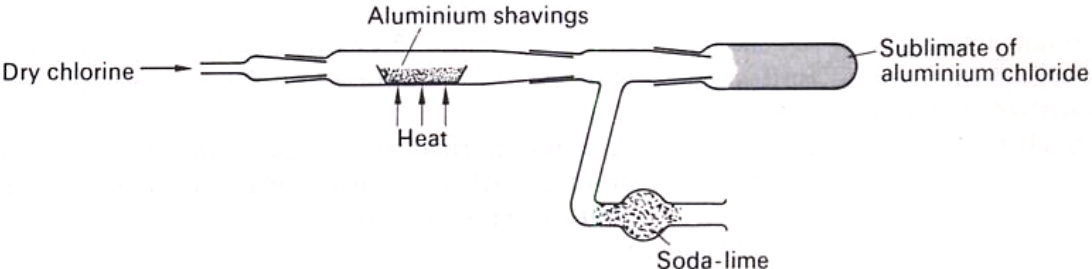
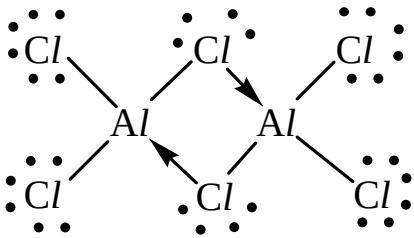
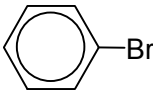
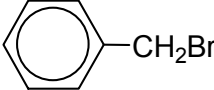
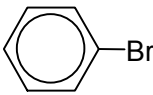
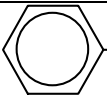


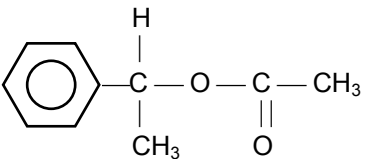
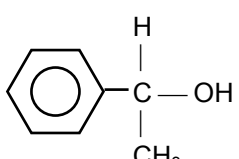
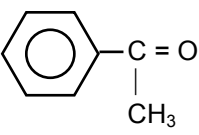
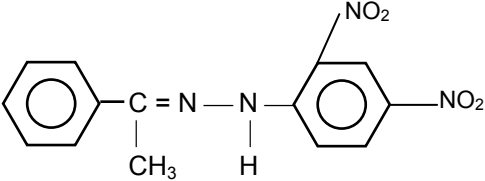
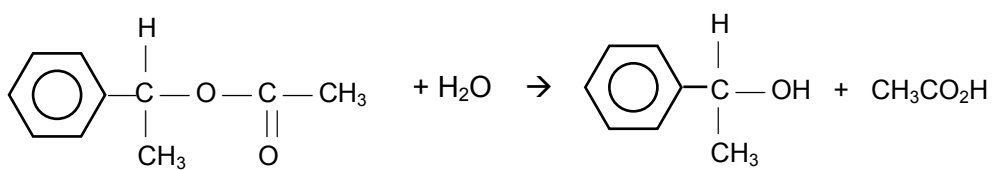
H1 Essay

5	(a)	The concentration of a hydrogen peroxide solution can be determined by titration with acidified potassium manganate(VII) solution. In this reaction the hydrogen peroxide is oxidised to oxygen gas. A 5.00cm³ sample of the hydrogen peroxide solution was added to a volumetric flask and made up to 250cm³ of aqueous solution. A 25.0 cm³ sample of this diluted solution was acidified and reacted completely with 24.35cm³ of 0.0187 mol dm⁻³ potassium manganate(VII) solution.
	(i)	Write an equation for the reaction between acidified potassium manganate(VII) solution and hydrogen peroxide.
		$2\text{MnO}_4^- + 6\text{H}^+ + 5\text{H}_2\text{O}_2 \longrightarrow 2\text{Mn}^{2+} + 8\text{H}_2\text{O} + 5\text{O}_2$
	(ii)	Use this equation and the results given to calculate a value for the concentration, in mol dm⁻³, of the original hydrogen peroxide solution. (If you have been unable to write an equation for this reaction you may assume that 3 mol of KMnO₄ react with 7mol of H₂O₂. This is not the correct reacting ratio.)
		Moles MnO₄⁻ = (24.35/1000) x 0.0187 = 4.55 x 10⁻⁴ Moles H₂O₂ = (4.55 x 10⁻⁴) x 5/2 = 1.138 x 10⁻³ Moles H₂O₂ in 5 cm³ original = (1.138 x 10⁻³) x 10 = 0.01138 Original [H₂O₂] = 0.01138 x (1000/5) = 2.28 mol dm⁻³
		If candidate uses given ratio 3/7, Original [H₂O₂] = 0.01062 x (1000/5) = 2.12 mol dm⁻³
	(b)	When dry chlorine is passed over heated aluminium foil in a hard glass tube, a vapour is produced which condenses to a yellow-white solid in the cooler parts of the tube. At low temperatures the vapour has the empirical formula AlCl₃ and M_r 267. 
	(i)	Suggest the molecular formula of the vapour, and draw a structure to describe its bonding.
		Molecular formula: Al₂Cl₆

			
		The yellow-white solid reacts with water in two separate ways. - When a few drops of water are added to the solid, steamy fumes are evolved and a white solid remains, which is insoluble in water. - When a large amount of water is added to the solid, a clear, weakly acidic solution results.	
		(ii)	Write the equations, including state symbols, for these two reactions. [6]
			A few drops of water: $2\text{AlCl}_3(\text{s}) + 3\text{H}_2\text{O}(\text{l}) \rightarrow \text{Al}_2\text{O}_3(\text{s}) + 6\text{HCl}(\text{g})$ OR $\text{AlCl}_3(\text{s}) + 3\text{H}_2\text{O}(\text{l}) \longrightarrow \text{Al}(\text{OH})_3(\text{s}) + 3\text{HCl}(\text{g})$ $\text{AlCl}_3(\text{s}) + 6\text{H}_2\text{O}(\text{l}) \rightarrow [\text{Al}(\text{H}_2\text{O})_5(\text{OH})]^{2+}(\text{aq}) + \text{H}^+(\text{aq}) + 3\text{Cl}^-(\text{aq})$
	(c)	The halogenoethanes $\text{C}_2\text{H}_5\text{Cl}$, $\text{C}_2\text{H}_5\text{Br}$ and $\text{C}_2\text{H}_5\text{I}$ differ in their physical properties and reactivities.	
		(i)	Suggest and explain how the boiling points of these compounds differ. The boiling point will increase, due to increasing van der Waals / id-id forces and larger no. of electrons/electron cloud or polarisation of clouds [‘not intermolecular forces’ on its own].
		(ii)	Describe and explain how the reactivities of the three compounds differ towards nucleophilic reagents. [3]
			Reactivity towards nucleophiles increases, Due to C–Hal bond strength / energy decreasing / weaker bonds
	(d)	One method of making 1-bromobutane in the laboratory is described below.	
		Stage 1	Place 35 g of powdered sodium bromide, 30 cm ³ of water, and 25 cm ³ (20 g) of butan-1-ol, in a 250 cm ³ two necked flask fitted with a tap funnel and reflux condenser.
		Stage 2	Concentrated sulfuric acid (25 cm ³) is then placed in the tap funnel

			and added drop by drop to the reagents in the flask, keeping the contents well shaken and cooled occasionally in an ice-water bath.
		The overall reaction may be considered to take place in two stages. In the first stage the two inorganic reagents react together to form HBr and a salt. In the second stage, the organic reagent reacts with the HBr that is formed in the first stage. Write an equation for each of these stages. [2]	
		stage I $\text{NaBr} + \text{H}_2\text{SO}_4 \rightarrow \text{NaHSO}_4 + \text{HBr}$ allow $2\text{NaBr} + \text{H}_2\text{SO}_4 \rightarrow \text{Na}_2\text{SO}_4 + 2\text{HBr}$ stage II $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH} + \text{HBr} \rightarrow \text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Br} + \text{H}_2\text{O}$	
	(e)	Chlorofluoroalkanes, CFCs, were once used as refrigerant fluids and aerosol propellants. In many applications they now have been replaced by alkanes. This is because CFCs contribute to the destruction of the ozone layer. Ultraviolet radiation from the sun in the stratosphere photolyses CFC molecules to form chlorine radicals, which initiate a chain reaction of destruction of ozone molecules.	
		(i)	Suggest one reason why CFCs were originally used for these purposes. They are <u>chemically and biologically inert</u> (and hence safe to use and handle), non-toxic and they can be <u>easily liquefied by pressure</u> a little above atmospheric pressure. They were also <u>volatilized readily</u> when pressure was released. Accept any of the above points.
		(ii)	Suggest one potential hazard of using alkanes instead of CFCs [2]
			The replacement of CFCs by alkanes has a danger because alkanes are <u>readily flammable</u>.
	(f)	Suggest simple test-tube reactions by which the isomers in the following pair can be distinguished from each other. You should state the reagents and conditions for each test, and describe how each of the isomers in the pair behaves.	
			 and  [2] [Total:20]
			Test: NaOH(aq) and heat. Add (excess) dilute HNO_3, followed by $\text{AgNO}_3\text{(aq)}$ Observation:  : cream precipitate of AgCl observed

			 : no cream precipitate of AgCl observed
6	(a)	The elements in Period 3 range from metals on the left of the Periodic Table to non-metals on the right. Using one property of the element and one of its chlorides, describe two pieces of evidence which support each of the following statements.	
		(i)	Sodium is a metal Na <u>conducts electricity</u> due to the presence of delocalized electrons as mobile charged carriers. Na loses its valence electron readily and forms an <u>ionic</u> chloride, Na^+Cl^- , which conducts electricity when molten (or in aqueous solution). <u>[Also accepted: Typical properties of metals. e.g. malleability. Typical properties of metallic salts e.g. forms solution in water, high boiling point.]</u>
		(ii)	Phosphorus is a non-metal Phosphorus, P_4 , is a simple molecule and does <u>not conduct electricity</u> at all. Phosphorus forms <u>covalent</u> chlorides, PCl_3 or PCl_5 , which reacts with water to form acidic solution. <u>[Also accepted: Typical properties of non-metals. e.g. low boiling point, dullness. Typical properties of PCl_3 or PCl_5. e.g. low boiling point, insulating properties, solubility in organic solvents.]</u>
			[4]
	(b)	State the shape and bond angle of the PF_3 molecule. Explain your answer using the Valence Shell Electron Pair Repulsion Theory.	
		PF_3 have <u>4 electron regions, 3 bond pairs and 1 lone pair</u> around the central atom. To minimize repulsion, the 4 electron pairs are directed to the <u>corners of a tetrahedron</u> with a <u>bond angle of 109.5°</u> . Since the repulsion between <u>lone pair-bond pair</u> > <u>bond pair-bond pair</u> , the bond angle is further compressed to <u>107°</u> and hence PF_3 is <u>trigonal pyramidal</u> .	
	(c)	A neutral liquid A , $\text{C}_{10}\text{H}_{12}\text{O}_2$ when heated with aqueous hydrochloric acid gave two products, B and C . B , $\text{C}_8\text{H}_{10}\text{O}$, gave a yellow precipitate when warmed with aqueous alkaline iodine. When heated with acidified sodium dichromate (VI), B gave D , $\text{C}_8\text{H}_8\text{O}$. D reacts with 2,4-dinitrophenylhydrazine to give orange crystals, E .	

		Deduce the possible structures of compounds A to E. Explain the chemistry of the reactions involved. Write an equation for A when it is heated with aqueous hydrochloric acid. [9]
		A  B  C CH₃CO₂H D  E  Equation:  A has a benzene ring as C:H ratio is almost 1:1. ° - A neutral liquid A, C₁₀H₁₂O₂ when heated with aqueous hydrochloric acid gave two products, B and C. → A is an <u>ester</u> and undergoes <u>acid hydrolysis</u> to form B and C. ° B gave a yellow precipitate when warmed with aqueous alkaline iodine. → B has <u>-CH₃CH(OH)</u> structure. ° - When heated with acidified sodium dichromate (VI), B gave D, C₈H₈O. → B undergoes <u>oxidation</u> to form a ketone, D. ° D undergoes condensation with 2,4-dinitrophenylhydrazine to give orange crystals, E. → D is a <u>ketone</u>. °
	(d)	How would you expect the acidity of 3-chloropropanoic acid and of fluoroethanoic acid to compare with that of chloroethanoic acid? [4] [Total: 20]
		Compare ClCH₂CH₂COOH vs ClCH₂COOH: ClCH₂CH₂COOH is <u>less acidic</u> than ClCH₂COOH because the electron withdrawing group, Cl atom is <u>further away from the carboxyl group and hence exerts a weaker electron withdrawing effect</u> in ClCH₂CH₂COOH than in ClCH₂COOH. The <u>negative</u>

		charge on O atom in $\text{ClCH}_2\text{CH}_2\text{COO}^-$ is less dispersed and the anion $\text{ClCH}_2\text{CH}_2\text{COO}^-$ is less stabilised. Compare FCH_2COOH vs ClCH_2COOH: FCH_2COOH is more acidic than ClCH_2COOH because electron-withdrawing effect F is greater than that of Cl due to greater electronegativity of F. The negative charge on O atom in FCH_2COO^- is more dispersed and the anion FCH_2COO^- is more stabilised. Acidity: $\text{FCH}_2\text{COOH} > \text{ClCH}_2\text{COOH} > \text{ClCH}_2\text{CH}_2\text{COOH}$
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7	(a)	The main buffering agent in blood plasma is the carbonic acid /hydrogen carbonate ion (H_2CO_3 / HCO_3^-) system. In blood plasma, the concentration of hydrogen carbonate ion is about twenty times the concentration of carbonic acid. The pH of arterial blood plasma is 7.40. If the pH falls below this normal value, a condition called <i>acidosis</i> is produced. If the pH rises above the normal value, the condition is called <i>alkalosis</i>. Write equations to show how the carbonic acid /hydrogen carbonate ion buffer system works to maintain the pH of blood plasma close to 7.40 on the addition of																								
	(i)	H^+ ions $\text{H}^+ + \text{HCO}_3^- \longrightarrow \text{H}_2\text{CO}_3$																								
	(ii)	OH^- ions $\text{OH}^- + \text{H}_2\text{CO}_3 \longrightarrow \text{HCO}_3^- + \text{H}_2\text{O}$																								
	(b)	Calculate the pH of a 0.30 mol dm^{-3} solution of sulfuric acid, H_2SO_4. $\text{H}_2\text{SO}_4 \longrightarrow 2\text{H}^+ + \text{SO}_4^{2-}$ $[\text{H}^+] = 2 \times 0.30 = 0.60 \text{ mol dm}^{-3}$ $\text{pH} = -\lg(0.60) = 0.222$																								
		[3]																								
	(c)	Propyl ethanoate undergoes a slow acid-catalysed hydrolysis in water. $\text{CH}_3\text{COOCH}_2\text{CH}_2\text{CH}_3 + \text{H}_2\text{O} \xrightarrow{\text{HCl}} \text{CH}_3\text{COOH} + \text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$ Two experiments were carried out, starting with different concentration of HCl. The following results were obtained. <table border="1"> <thead> <tr> <th>time / minutes</th><th>Experiment 1, with $[\text{HCl}] = 0.10 \text{ mol dm}^{-3}$ $[\text{CH}_3\text{COOCH}_2\text{CH}_2\text{CH}_3] / \text{mol dm}^{-3}$</th><th>Experiment 2, with $[\text{HCl}] = 0.05 \text{ mol dm}^{-3}$ $\text{CH}_3\text{COOCH}_2\text{CH}_2\text{CH}_3 / \text{mol dm}^{-3}$</th></tr> </thead> <tbody> <tr><td>0</td><td>0.0050</td><td>0.0050</td></tr> <tr><td>15</td><td>0.0040</td><td>0.0045</td></tr> <tr><td>30</td><td>0.0032</td><td>0.0040</td></tr> <tr><td>45</td><td>0.0026</td><td>0.0036</td></tr> <tr><td>60</td><td>0.0021</td><td>0.0032</td></tr> <tr><td>75</td><td>0.0017</td><td>0.0029</td></tr> <tr><td>90</td><td>0.0014</td><td>0.0026</td></tr> </tbody> </table>	time / minutes	Experiment 1, with $[\text{HCl}] = 0.10 \text{ mol dm}^{-3}$ $[\text{CH}_3\text{COOCH}_2\text{CH}_2\text{CH}_3] / \text{mol dm}^{-3}$	Experiment 2, with $[\text{HCl}] = 0.05 \text{ mol dm}^{-3}$ $\text{CH}_3\text{COOCH}_2\text{CH}_2\text{CH}_3 / \text{mol dm}^{-3}$	0	0.0050	0.0050	15	0.0040	0.0045	30	0.0032	0.0040	45	0.0026	0.0036	60	0.0021	0.0032	75	0.0017	0.0029	90	0.0014	0.0026
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		(i)	Using the same axes, plot graphs of $[\text{CH}_3\text{COOCH}_2\text{CH}_2\text{CH}_3]$ against time for the two experiments.
			propyl ethanoate/mol dm^{-3} time / min Experiment 1 (pink squares), Experiment 2 (blue diamonds)
		(ii)	Use your graphs to determine the order of reaction with respect to $[\text{CH}_3\text{COOCH}_2\text{CH}_2\text{CH}_3]$ and to $[\text{HCl}]$.
			When $[\text{HCl}] = 0.10 \text{ mol dm}^{-3}$ Initial rate for experiment 1 = $\frac{0.005}{60} = 8.33 \times 10^{-5}$ When $[\text{HCl}] = 0.05 \text{ mol dm}^{-3}$ Initial rate for experiment 2 = $\frac{0.005}{120} = 4.17 \times 10^{-5}$ When $[\text{HCl}]$ is halved, the initial rate is also halved. Hence the rate of reaction is <u>first order</u> with respect to HCl. Using Experiment 1 Either When [propyl ethanoate] reduce from 0.0050 to 0.0025 mol dm^{-3} $t_{1/2} = 50 \text{ s}$ When [propyl ethanoate] reduce from 0.0040 to 0.0020 mol dm^{-3} $t_{1/2} = 51 \text{ s}$ or half lives are indicated (with proper lines) on the graph Since the two $t_{1/2}$ is <u>relatively constant</u>, the reaction is <u>first order</u> with respect to propyl ethanoate.

		(iii)	Construct a rate equation for the reaction and use it to calculate a value for the rate constant, giving its units.												
			$\text{Rate} = k[\text{CH}_3\text{COOCH}_2\text{CH}_2\text{CH}_3][\text{HCl}]$ $k = \frac{8.33 \times 10^{-5}}{(0.10)(0.0050)} = 0.167 \text{ mol}^{-1} \text{ dm}^3 \text{ min}^{-1}$												
			[9]												
	(d)		At room temperature, gaseous dinitrogen tetraoxide and nitrogen dioxide are in dynamic equilibrium according to the following equation: $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g}) \qquad K_c = 7.246 \times 10^{-3}$ Colourless brown The mixture turns into a darker shade of brown when it was placed in hot water.												
		(i)	Deduce and explain whether the forward reaction is endothermic or exothermic.												
			At higher temperature, position of equilibrium shifts to the right since mixture turns into a darker shade of brown. Since increase in temperature favours the endothermic reaction to absorb additional heat, forward reaction is endothermic.												
		(ii)	Explain, with reason, whether the dissociation of N_2O_4 is favoured by high or low pressure.												
			It is favoured at low pressure. At low pressure, position of equilibrium shifts to the right where there are more gaseous molecules to increase the pressure.												
		(iii)	1.38 g of N_2O_4 was placed in a 2.0dm^3 reaction vessel. At equilibrium, it was found that N_2O_4 was 40% dissociated. Calculate the equilibrium concentration of NO_2 .												
			Initial mol of $\text{N}_2\text{O}_4 = \frac{1.38}{92.0} = 0.015 \text{ mol}$ <table><tr><td></td><td>$\text{N}_2\text{O}_4(\text{g})$</td><td>$2\text{NO}_2(\text{g})$</td></tr><tr><td>Initial / mol</td><td>0.015</td><td>0</td></tr><tr><td>Change / mol</td><td>-0.006</td><td>+0.012</td></tr><tr><td>Equilibrium / mol</td><td>0.009</td><td>0.012</td></tr></table> $[\text{NO}_2] = 0.00600 \text{ mol dm}^{-3}$		$\text{N}_2\text{O}_4(\text{g})$	$2\text{NO}_2(\text{g})$	Initial / mol	0.015	0	Change / mol	-0.006	+0.012	Equilibrium / mol	0.009	0.012
	$\text{N}_2\text{O}_4(\text{g})$	$2\text{NO}_2(\text{g})$													
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Change / mol	-0.006	+0.012													
Equilibrium / mol	0.009	0.012													

		(iv)	Draw a labelled reaction pathway diagram for the dissociation of N_2O_4 .
			[8]
			[Total:20]